

# Generative AI and Social Media



**Tauhid Zaman**

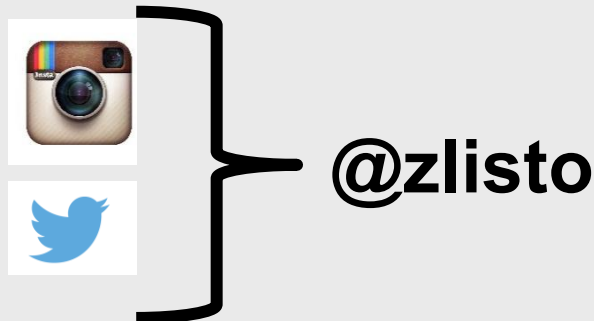
# **Generative AI and Social Media**

- **Learn to use AI**
  - **For analyzing and creating social media data**
  - **For building social media related web and mobile applications**
- **No coding experience required**
  - **AI does most of the work for us**



# Bio - Tauhid Zaman

- Associate Professor and Special Advisor for AI Initiatives at Yale School of Management
- BS, MEng, PhD in Electrical Engineering and Computer Science from MIT
- Research: AI, social media, cryptocurrency, sports analytics
- Email: [tauhid.zaman@yale.edu](mailto:tauhid.zaman@yale.edu)

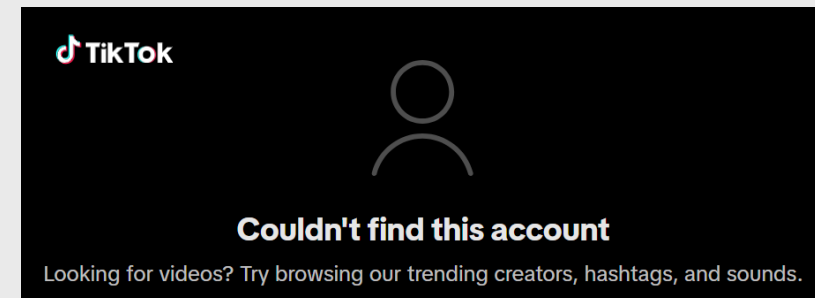
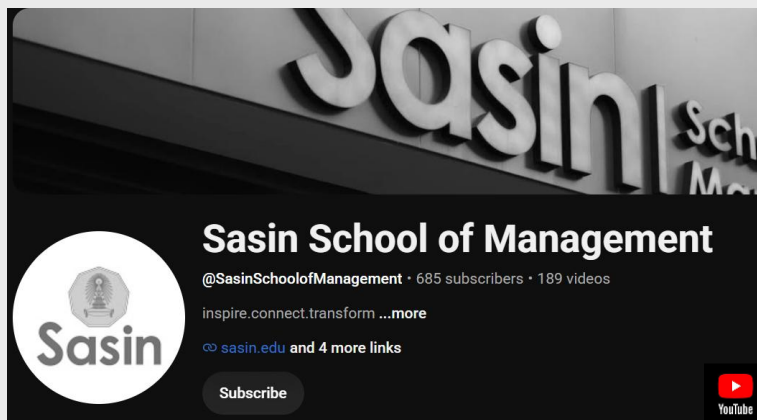


# Course Info

- **Course website:**  
[https://zlisto.github.io/genAI social media 2026 summer](https://zlisto.github.io/genAI_social_media_2026_summer)



# What Can We Do With AI + Social Media?



# Learning About an Audience

**Search engines**



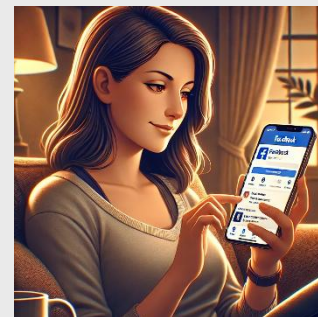
**What you need**



**Social media**



**What you like**



# Learning About an Audience

## AI Agents



## What you feel



# AI Ads

## ChatGPT is now doing what Sam Altman called a 'last resort' only two years ago

After vowing not to, OpenAI's CEO is forced to backtrack

By [Jules Rimmer](#) [Follow](#)

Last Updated: Jan. 20, 2026 at 3:17 a.m. ET

First Published: Jan. 19, 2026 at 6:52 a.m. ET

 Share

 Resize

 9

 Listen (4 min)





# Our approach to advertising and expanding access to ChatGPT

 Listen to article | 4:36

 Share

AI is reaching a point where everyone can have a personal super-assistant that helps them learn and do almost anything. Who gets access to that level of intelligence will shape whether AI expands opportunity or reinforces the same divides.

We've been working to make powerful AI accessible to everyone through our free product and low-cost subscription tier, ChatGPT Go, which has launched in 171 countries since August. Today we're bringing Go to the U.S. and everywhere ChatGPT is available, giving people expanded access to messaging, image creation, file uploads and memory for \$8 USD/month. In the coming weeks, we're also planning to start testing ads in the U.S. for the free and Go tiers, so more people can benefit from our tools with fewer usage limits or without having to pay. Plus, Pro, Business, and Enterprise subscriptions will not include ads.



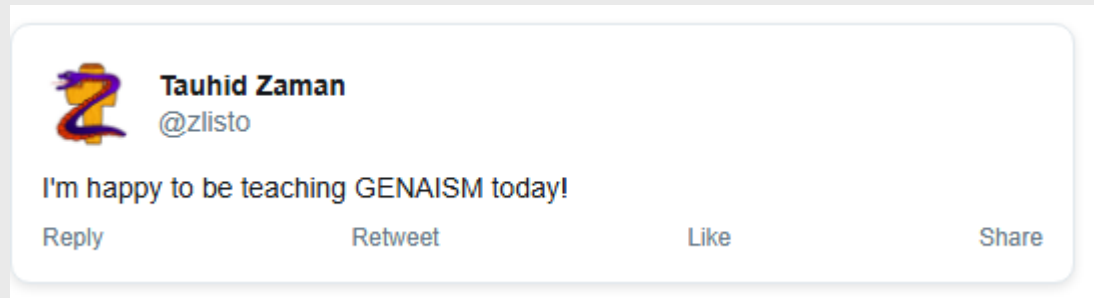
# **Analyzing Social Media Data with AI**

- **Measure sentiment**
- **Community detection**
- **Image analysis**



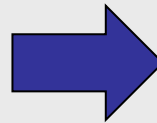
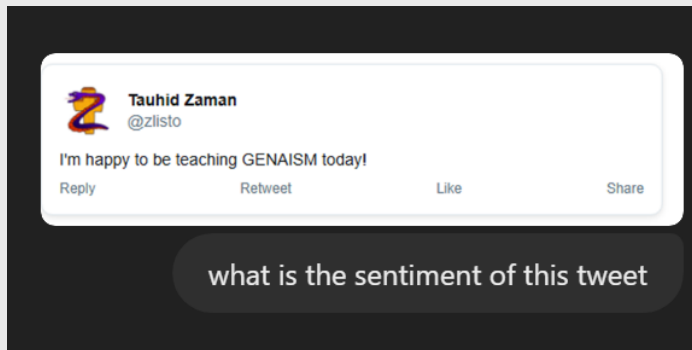
# Measuring Sentiment

- **Old way: train a model that looks for happy or sad words in text**



# Measuring Sentiment with AI

- New way: ask AI



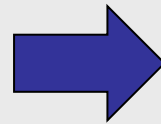
## Sentiment Analysis:

- **Polarity:** 1.0 (Positive sentiment)
- **Subjectivity:** 1.0 (Completely subjective)

The sentiment is highly positive, reflecting happiness and enthusiasm. [:-]

# Community Detection

- Old way: use network structure and machine learning algorithms



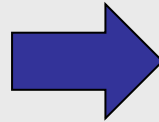
# Community Detection with AI

- New way: ask AI

TwFollow Joe Biden Following

|   | name            | username   | userId       |
|---|-----------------|------------|--------------|
| 1 | President Biden | POTUS      | '1349149363' |
| 2 | The White House | WhiteHouse | '1323730784' |

Put these people into communities based on their twitter bios

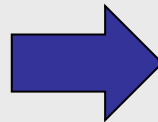
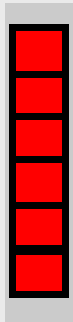
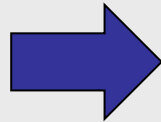


Social Network Communities

| Name     | Community         |
|----------|-------------------|
| Alex     | Friends from Work |
| Sophia   | Friends from Work |
| James    | College Friends   |
| Emma     | College Friends   |
| Liam     | Neighborhood      |
| Olivia   | Neighborhood      |
| Noah     | Family            |
| Ava      | Family            |
| William  | Online Friends    |
| Isabella | Online Friends    |

# Image Analysis

- Old way: train a classifier on labeled images



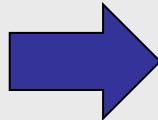
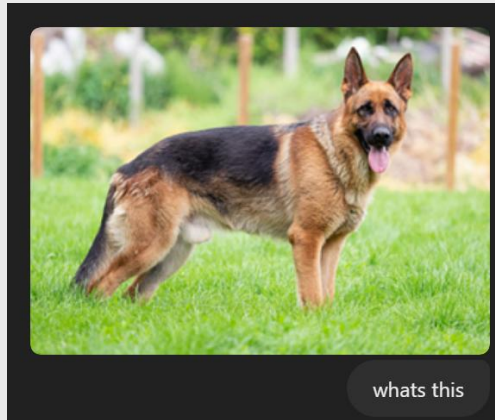
What is this a picture of?

- ☒ Dog
- ☐ Cat
- ☐ Plane



# Image Analysis with AI

- New way: ask AI



This image appears to show a German Shepherd dog. Let me know if you need further details or analysis!





# **Creating Social Media Content with AI**

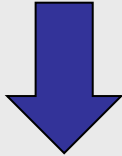
- **Text content**
- **Image content**
- **Audio content**
- **Video content**



# Text Content

## Tweet

write a tweet about how hyped i am to be teaching Generative AI and Social Media (GENAISM) and put in a funny joke about Elon Musk.



**Tauhid Zaman**  
@zlisto

Super hyped to be teaching Generative AI and Social Media (GENAISM)! 🚀 If this goes well, I might just rebrand myself as 'Professor X.AI' and wait for Elon Musk to send me a cease-and-desist. 🤪 #GENAISM #AI

Reply

Retweet

Like

Share

## Image Caption



write a funny tweet about this image and put in a funny joke about Elon Musk



**Tauhid Zaman**  
@zlisto

Breaking: Elon Musk unveils the next-gen Tesla DogBot. It doesn't fetch, but it might tweet back at you. Beta testing begins in your backyard. 🐕🤖 #DogeCoin #Innovation

Reply

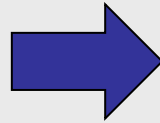
Retweet

Like

Share

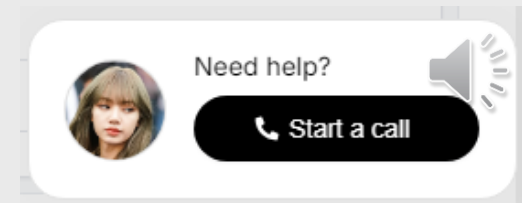
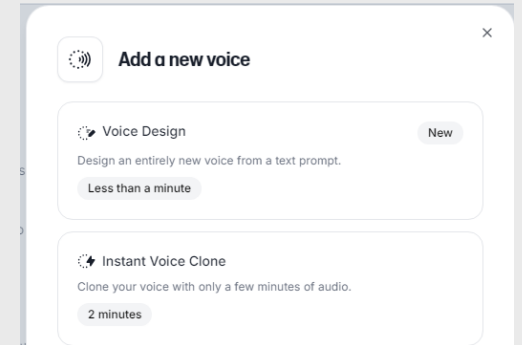
# Image Content

draw a catchy Instagram image for this tweet:  
Breaking: Elon Musk unveils the next-gen Tesla  
DogBot. It doesn't fetch, but it might tweet  
back at you. Beta testing begins in your  
backyard. 🐕🤖 #DogeCoin #Innovation



# Audio Content

- **AI voices**
  - AI can easily convert text to speech
  - AI can also clone someone's voice
  - Allows us to make narrations for videos and conversational agents
- **AI music**
  - AI can create songs from text descriptions
  - Any musical genre
  - Any language



# Video Content

- **AI video generation from an idea**

write me a video script (scene and narration) for a video telling people to study at Yale SOM because our mascot is a cool bulldog





# Building Apps with AI



# Building Apps with AI

- In this course you will build web applications for social media
- To build a web app you need to know
  - HTML
  - JavaScript
  - Python
  - SQL
- In the old days you'd hire a team of software developers to build your app for you
  - Takes a long time 
  - Pay them a lot \$\$



# Vibe Coding

- **Vibe coding – tell the AI the vibe of your app and it writes the code for it and runs it**
- **Once you have the basics down, you can make some very creative apps**
- **Some examples**
  - Behavioral analyst
  - Video narration generator
  - Movie script writer
  - Content creator agent
  - And many more ...



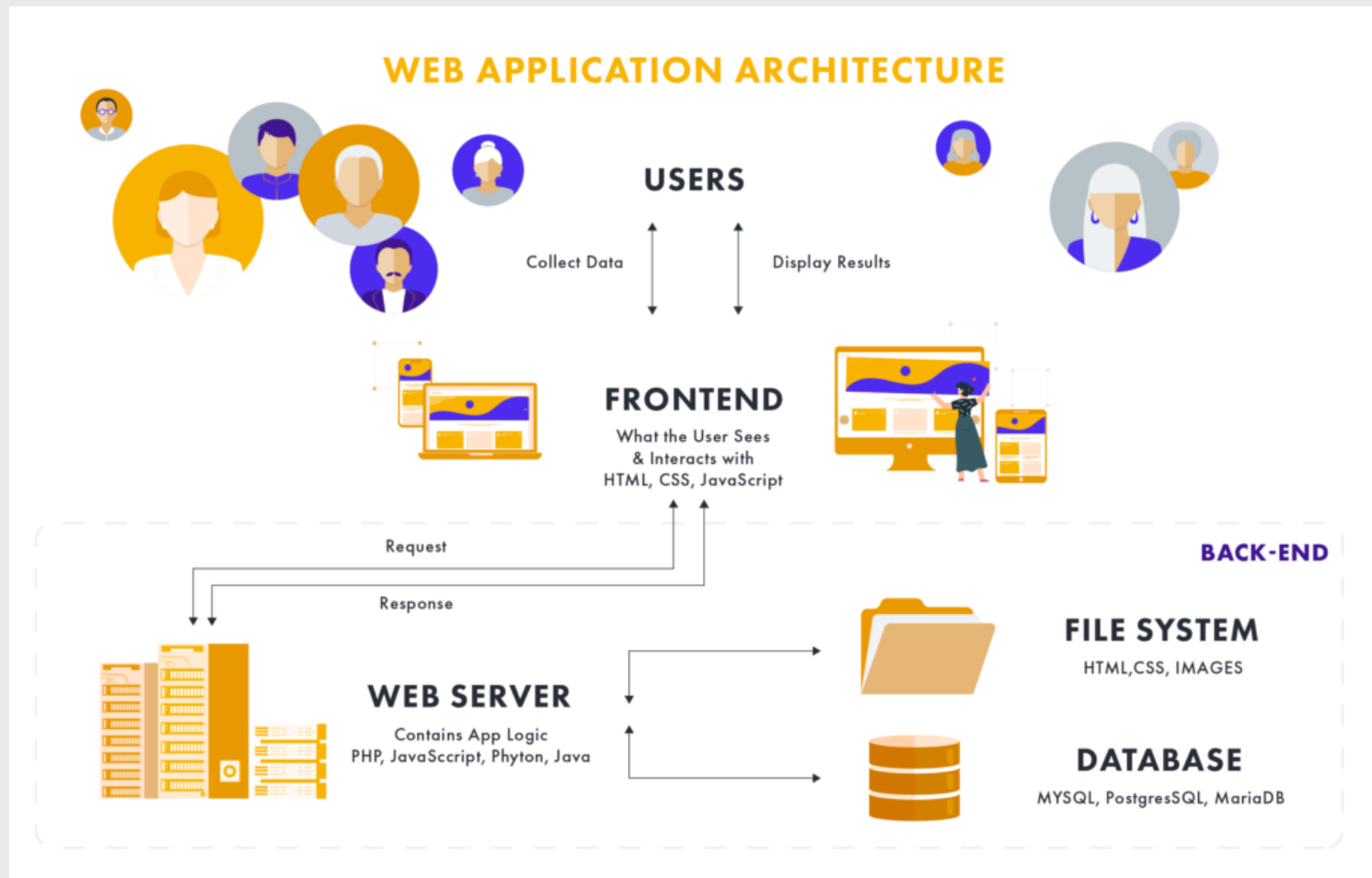


# Types of Apps

- **Web apps**
  - Run on a web browser
  - Written in HTML, CSS, Javascript, Python
  - Built with frameworks like React or Streamlit
  - Easy to deploy
- **Mobile apps**
  - Run on a mobile phone
  - Written in Swift (iOS) or Kotlin (Android)
  - Frameworks like React Native or Expo Go
  - Require more work to deploy (but not too much more)



# App Structure



# Python & Streamlit



- **Streamlit is a Python web app framework**
  - You will need to install Python to use it
- **Pros**
  - Easy to get started
  - No need to write frontend code (everything is Python)
- **Cons**
  - Not flexible
  - Front end doesn't look great
- **This week we will use Streamlit to get started**
  - Later we will use much better frameworks like React



# Example Streamlit App

## Twitter User Analysis Dashboard

Upload a CSV file with tweets



Drag and drop file here

Limit 200MB per file • CSV

Browse files



TwExportly\_Cobratate\_tweets\_2024\_04\_19.csv 330.9KB



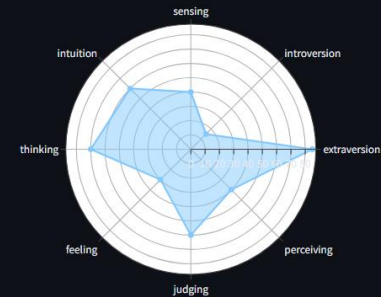
✓ Analyzing tweets... 🔄

✓ Analyzing personality... 🔄

## Topics

| topic                                       | summary                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wealth and Personal Responsibility          | The user emphasizes the importance of personal effort and responsibility in achieving wealth and success. They argue against complacency and stress that individuals should not rely on external circumstances or blame but take charge of their lives. Wealth is framed as not just a material possession but an avenue for freedom and influence in one's life.                                                                         |
| Critique of Modern Society and Institutions | There are strong criticisms of various aspects of modern society, including government actions, media manipulations, and sociocultural norms that are seen as detrimental to true progress. The user believes that significant issues like inflation, societal collapse, and the erosion of traditional values stem from the failure of 'Western democracy' and the influence of 'The Matrix' (a metaphor for systemic societal control). |
| Charity and Community Support               | The user frequently mentions charitable activities and the importance of helping those in need. They express a strong commitment to philanthropy and view it as a responsibility of successful individuals. This is framed as a way to not only aid the less fortunate but also to reflect one's character and values in the community.                                                                                                   |

## Personality Traits Radar Plot



## Persona Chat



11:15:38: hey whats up



Hey, it's Andrew. Just living the life, focusing on turning ambitions into reality. What about you?



11:15:58: tell me about the Matrix



✓ AI is thinking... 🔄



11:16:03: The Matrix is all about the system that keeps most people blind to the truth. It's the societal constructs, the lies we're fed that keep us in a state of complacency. Most people are

Type your message...



# Vibe Coding an App

- **App prompt - basic idea and flow of app**
  - Text description
  - Image of a flow diagram
- **App code – written by an AI vibe coder**
  - Cursor, Claude Code, etc.
- **Run the app**
  - Run it yourself from the terminal
  - Tell the AI vibe coder to run it for you



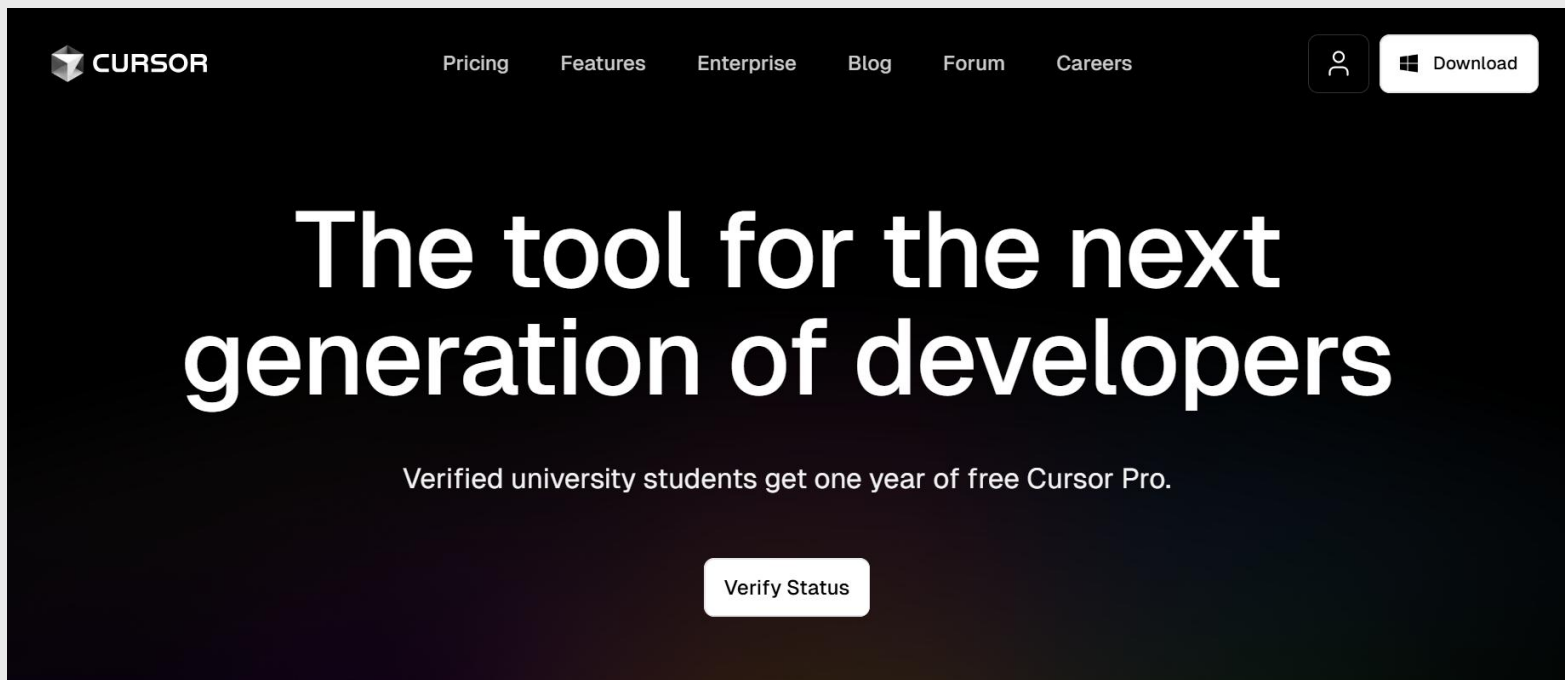
# Vibe Coders

- **Today there are many AI-powered vibe coders**
  - Require (almost) zero coding knowledge
  - Cost money \$\$\$ (minimum \$20 per month, but can go higher depending on use)
  - Provide an AI agent that writes and executes your code
- **These are basically chatbots attached to clever scaffolds and loops**



# Cursor

- **We will build apps in Cursor**
  - **Used to be free for students for one year (need a .edu email address)**

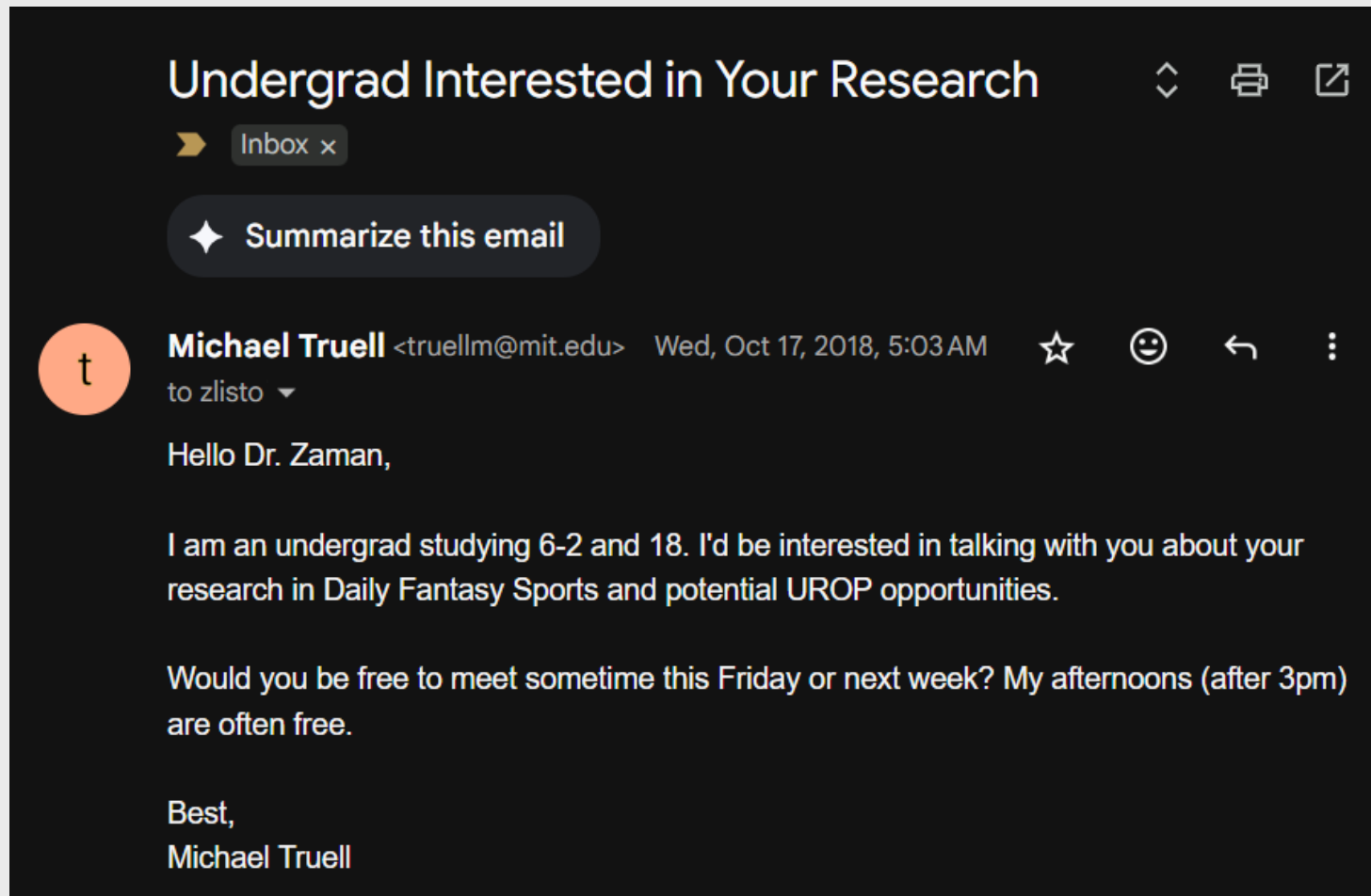


# Cursor Founder





# Before He Founded Cursor



# Cursor Growth

**SpaceX to acquire the AI coding startup Cursor for \$60 billion**

PUBLISHED TUE, JUN 16 2026 9:06 AM EDT | UPDATED TUE, JUN 16 2026 10:51 AM EDT



# Cursor Interface

The image displays the Cursor IDE interface with the following components:

- File directory:** Located on the left, it shows a project structure with files like `corr_trade_simulator.py` and `corr_trading.ipynb`.
- Code editor:** The central area displays the `corr_trade_simulator.py` file, showing a Python class `CorrTraderSimulator` with methods `compute_tstat` and `compute_df_tstat`.
- Terminal:** At the bottom, it shows the output of the code execution, including data loading and trading simulation details.
- AI Chat:** On the right, it provides a chat interface for asking questions or getting assistance with the code.

Labels are placed over the interface to identify these sections:

- File directory:** Points to the left sidebar.
- Code editor:** Points to the central code area.
- Terminal:** Points to the bottom output area.
- AI Chat:** Points to the right sidebar chat area.

# Cursor Installation

- **Steps for setting up Cursor for making Python based apps**
  1. **Sign up for Cursor with your .edu email address**
  2. **Download and install Cursor**
  3. **Download and install Python**
  4. **Connect Cursor to Python. Cursor needs to know where your Python is on your computer**
- **Detailed instructions on course website on [Instructions](#) page and on this [YouTube video](#)**



# **COURSE LOGISTICS**



# Grading

- **Grading:**
  - **Homework Assignments: 40%**
  - **Final Project Report: 60%**



# Homework Assignments

- Weekly homework assignments (usually)
- Homework submissions are generally Python code files (.py)
- Should be completed **individually** (AI-maxing is encouraged)
- Late homework will not be accepted (unless University approves it) so don't bother asking



# Final Project

- **Topic – Something that involves the novel analysis or creation of social media data**
- **Examples**
  - Build a web app that creates virtual podcasts between celebrities on a topic of your choice
- **Teams must be formed by July 20<sup>th</sup>**
- **Topics chosen by August 3<sup>rd</sup>**
- **Final presentation during last day of class**
- **Final report due on last day of class**





# Final Project Teams

1 person



2 people



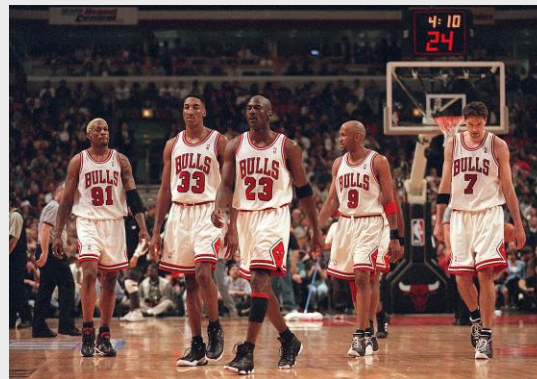
3 people



4 people



5 people



6 people



# Next Class: Coding Session

- We will build an app to measure complex tweet sentiment with AI

You will be given a sentence and must grade its Ekman's Six Basic Emotions from 0 to 100: Return a table of the scores and explanation why: I hate my job

Here is an analysis of the sentence "I hate my job," graded on Ekman's Six Basic Emotions (Anger, Fear, Joy, Sadness, Disgust, and Surprise), along with explanations for each score:

| Emotion  | Score (0-100) | Explanation                                                                                              |
|----------|---------------|----------------------------------------------------------------------------------------------------------|
| Anger    | 85            | The word "hate" conveys strong negative feelings and frustration, indicating anger.                      |
| Fear     | 40            | The sentiment may imply fear of being stuck in an undesirable situation or uncertainty about the future. |
| Joy      | 0             | The sentence expresses no happiness or positivity, as "hate" and "job" in this context are negative.     |
| Sadness  | 75            | The phrase suggests unhappiness, disappointment, or distress about the situation.                        |
| Disgust  | 90            | The word "hate" evokes a sense of aversion or repulsion, strongly tied to disgust.                       |
| Surprise | 10            | There is little indication of surprise in the sentence, as it reflects ongoing dissatisfaction.          |



# Before Next Class

- Install Cursor and Python (instructions in YouTube video or course website)
- Get ready to vibe code

